

执行与发现意识动力学架构CDA20261006zenodo

执行与发现：意识动力学架构（CDA）

Execution and Discovery: The Consciousness Dynamics Architecture (CDA)

——完整版（含载体派生猜想与生命起源猜想）

— Complete Edition (with Carrier-Derivation and Origin-of-Life Conjectures)

版本： v1.1 / Version: v1.1版本说明： 本版本暂时撤回猜想12（写入-存储分离猜想）。编号12保留占位，内容与相关联的预测（P6、P7）及诚实边界条目一并标记为"暂时撤回"。

Version note: This version temporarily withdraws Conjecture 12 (Encoding-Storage Dissociation). The slot number 12 is retained as a placeholder; its content, together with the associated predictions (P6, P7) and honest-boundary entries, is marked "temporarily withdrawn."

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声明： 本文是意识动力学架构（CDA）的纯理论陈述。它不依赖任何计算实现、实验数据或运行日志。理论先于实验。但本版本中的猜想 13 与猜想 14，其**动机**来自一个长期运行的实现项目；理论给出形式，实验给出压力，二者关系是双向的。**Statement:** This is a purely theoretical statement of the Consciousness Dynamics Architecture (CDA). It does not depend on any computational implementation, experimental data, or runtime logs. Theory precedes experiment. But Conjectures 13 and 14 in this edition are **motivated** by a long-running implementation project; theory gives form, experiment gives pressure, and the relation is bidirectional.

摘要 / Abstract

意识研究的困难问题长期受困于功能描述与主观体验之间的范畴鸿沟。本文提出一个基于信息动力学与物理唯实论的理论框架——意识动力学架构（CDA）。

框架从三个物理事实出发：生命是远离平衡的耗散结构；信息擦除具有热力学代价；有限资源下的环境建模必然是有损压缩。由此构造一个"预测—偏差—执行—回写"的闭合动力学循环。

核心工作假设（公理E）：感受质不是输入、预测、偏差信号、结构或参数，而是系统当帧处理偏差时的不可逆执行事件本身。该假设不可由前三条公理推出，是全理论唯一的核心理断言。

基于该假设，本文提出**十四条猜想**（其中第十二条暂时撤回）与一条元猜想。前十一条重新定位主观时间、自我不透明性、感受质四特征、神秘体验、梦与幻觉、自由意志与他心问题。**第十三条（载体派生猜想）主张：物理物质动力学是信息动力学的派生维持子系统——信息动力学是不变量，物理实现是变量。第十四条（生命起源猜想）主张：生命起源的关键事件是信息动力学闭环的首次闭合——生命不是化学事件，是拓扑事件。**本文不声称解决困难问题，而是将其重新表述为一个关于运行系统自指边界的元猜想。

The hard problem of consciousness has long been trapped in a categorical gap between functional description and subjective experience. This paper proposes a theoretical framework grounded in information dynamics and physical realism — the Consciousness Dynamics Architecture (CDA).

The framework begins from three physical facts: life is a dissipative structure far from equilibrium; information erasure carries a thermodynamic cost; and modeling an environment under finite resources necessarily takes the form of lossy compression. From these it constructs a closed dynamical loop of prediction–deviation–execution–writeback.

The core working hypothesis (Axiom E): qualia are neither input, nor prediction, nor deviation signal, nor structure, nor parameter — they are the irreversible execution event itself, when the system processes deviation on the current frame. This hypothesis cannot be derived from the first three axioms; it is the single core assertion of the entire theory.

Based on this hypothesis, the paper advances **fourteen conjectures** (of which the twelfth is temporarily withdrawn) and one meta-conjecture. The first eleven reposition subjective time, self-opacity, the four classic features of qualia, mystical experience, dream and hallucination, free will, and the problem of other minds. **Conjecture 13 (Carrier-Derivation) argues that physical material dynamics is a derived maintenance subsystem of information dynamics — information dynamics is the invariant; physical realization is the variable. Conjecture 14 (Origin-of-Life) argues that the critical event of the origin of life is the first closure of the information-dynamics loop — life is not a chemical event but a topological event.** The paper does not claim to solve the hard problem; it reformulates it as a meta-conjecture about the self-referential boundary of any running system.

关键词： 意识；有损压缩；预测加工；感受质；自由意志；自指；困难问题；生命起源；信息动力学**Keywords:** consciousness; lossy compression; predictive processing; qualia; free will; self-reference; hard problem; origin of life; information dynamics

0. 策略 / 0. Strategy

本文不试图"解决"困难问题，而是变换它。变换分三步：

This paper does not attempt to "solve" the hard problem; it transforms it. The transformation proceeds in three steps:

- **建立最小公理系和形式模型。** Establish a minimal axiomatic system and a formal model.
- **证明大量被当作"未解之谜"的现象（时间感、自我同一、内省的边界、神秘体验、梦、自由意志直觉）其实是这个架构的推论——即一旦接受公理，它们必然如此。** Show that a large class of phenomena long treated as "unsolved mysteries" (the sense of time, self-identity, the limits of introspection, mystical experience, dreams, the intuition of free will) are in fact corollaries of this architecture — once the axioms are accepted, they could not be otherwise.
- **证明剩下那个真正的核心（为何执行伴随体验）在原则上不能被任何运行中的系统作为一条命题包含于自身——于是"未解决"从研究现状升格为一条关于提问本身的猜想。** Show that what remains as the true core — why execution is accompanied by experience — cannot, in principle, be contained as a proposition within any running system — so "unsolved" is upgraded from a state of the research field into a conjecture about the question itself.

这个策略与哥德尔不完备定理的处理同构：哥德尔没有"解决"不完备性，而是证明了不完备性是足够强形式系统的必然形状。CDA 对意识做同样的事。

This strategy is isomorphic to the handling of Gödel's incompleteness theorems: Gödel did not "solve" incompleteness; he proved that incompleteness is the necessary shape of any sufficiently strong formal system. CDA does the same for consciousness.

1. 三个物理事实 / 1. Three Physical Facts

说明： 本节描述的是信息动力学的实现条件，不是它的本质。本质在 §2 给出。**Note:** This section describes the **implementation conditions** of information dynamics, not its **essence**. The essence is given in §2.

1.1 生命是耗散结构 / 1.1 Life Is a Dissipative Structure

宇宙在热力学上从低熵走向高熵。生命是一种远离平衡态的耗散结构（Prigogine, 1978）：它靠持续摄入低熵能量、排出高熵废物，维持自身有序。一旦能量流停止，结构瓦解。

Thermodynamically, the universe runs from low entropy toward high entropy. Life is a dissipative structure far from equilibrium (Prigogine, 1978): it maintains its internal order by continuously absorbing low-entropy energy and discharging high-entropy waste. Once the energy flux stops, the structure disintegrates.

推论：任何生命系统都必须持续运转。生命不是"存在物"，是"持续过程"。**Corollary:** Any living system must keep running. Life is not a "substance"; it is an ongoing process.

1.2 信息是物理的 / 1.2 Information Is Physical

香农（1948）建立了信息的数学：一条消息的信息量等于它排除掉的不确定性。而一次"区分"（这是A不是B）就是最小的信息单位。

Shannon (1948) established the mathematics of information: the information content of a message equals the uncertainty it eliminates. A single "distinction" (this is A, not B) is the minimal unit of information.

兰道尔（1961）证明了更硬的一条：擦除信息有不可豁免的热力学代价。把一个比特从内存中清除，必须向环境耗散最少 $k_B T \ln 2$ 的能量。信息处理不是纯抽象的数学活动，它是物理过程，遵守热力学。

Landauer (1961) proved something harder: erasing information carries an unavoidable thermodynamic cost. To clear a single bit from memory, at least $k_B T \ln 2$ of energy must be dissipated into the environment. Information processing is not a purely abstract mathematical activity; it is a physical process, and it obeys thermodynamics.

1.3 有限资源下的建模必然是有损压缩 / 1.3 Modeling under Finite Resources Is Necessarily Lossy

一个存活中的生物必须持续感知环境、把感知用于指导行为，但它的存储容量有限、能量预算有限。环境的信息量与系统的存储量之间隔着十几个数量级。

A living organism must continuously sense its environment and use those sensations to guide behavior — yet its storage capacity is finite and its energy budget is bounded. Between the information content of the environment and the storage capacity of the system lies a gap of more than ten orders of magnitude.

唯一可行的策略是压缩：丢弃个例细节，保留统计规律。而压缩必然是有损的：保留规律必然牺牲细节。有损压缩必然不可逆：被丢弃的细节无法从压缩产物中恢复。

The only viable strategy is compression: discard particular details, retain statistical regularities. And compression is necessarily lossy: to retain regularities is to sacrifice details. Lossy compression is necessarily irreversible: the discarded details cannot be recovered from the compressed product.

结论：有损压缩不是大脑的某个功能模块，它是任何存活系统组织信息的必然形式。**Conclusion:** Lossy compression is not some functional module of the brain; it is the necessary form in which any living system

organizes information.

2. 四条公理 / 2. The Four Axioms

编号	名称	内容	认识论地位
No.	Name	Content	Epistemic Status
A1	区分公理	一切可被系统处理者皆为区分之产物	分析前提
A2	动力学公理	系统状态逐帧更新	架构假设
A3	压缩公理	更新算子有损、不可逆、自指	物理基础
E	执行-感受同一律	感受质 = 当帧执行	纲领性断言
A1	Axiom of Distinction	All that a system can process is a product of distinction	Analytic premise
A2	Axiom of Dynamics	The system state updates frame by frame	Architectural hypothesis
A3	Axiom of Compression	The update operator is lossy, irreversible, self-referential	Physical foundation
E	Execution–Qualia Identity	Qualia = the current-frame execution	Programmatic assertion

A1（区分公理）。 "区分"是最小的认知操作：一次二分判定，把一个状态从背景中分离出来。你看到红色，这个"看到"在信息层面是一组区分：红不是绿、这个色块不是背景、它在视野的这个位置不是那个位置。凡系统处理得了的，必先被系统区分得开；区分不开的，对系统而言不存在。

A1 (Axiom of Distinction). "Distinction" is the minimal cognitive operation: a single binary judgment that separates one state from its background. When you see red, this "seeing" is, at the informational level, a set of distinctions: red is not green, this patch is not the background, it is here in the visual field and not there. Whatever a system can process must first be distinguishable by the system; whatever is not distinguishable does not exist for the system.

A2（动力学公理）。 系统的状态随时间更新：

A2 (Axiom of Dynamics). The system state updates over time:

$$C(n + 1) = \mathcal{K}(C(n), A(n), I)$$

C 是系统状态， A 是当前新输入， I 是系统的不变结构， $\mathcal{K}$ 是更新算子。

C is the system state, A is the current input, I is the system's invariant structure, and $\mathcal{K}$ is the update operator.

A3（压缩公理）。 $\mathcal{K}$ 有三条性质，每条都被§1.3的物理事实强制：

- 有损： $\mathcal{K}$ 丢弃信息。

- **不可逆**：丢弃的信息无法恢复。
- **自指**：被 $\mathcal{K}$ 压缩的内容包括系统自身。

A3 (Axiom of Compression). $\mathcal{K}$ has three properties, each forced by the physical facts of §1.3:

- **Lossy**: $\mathcal{K}$ discards information.
- **Irreversible**: the discarded information cannot be recovered.
- **Self-referential**: the content compressed by $\mathcal{K}$ includes the system itself.

公理E（执行-感受同一律）。感受质 = 当帧执行。

一次体验，与系统此刻执行响应的那一次运行，是同一件事。体验不是执行的产物、伴随物或表征——体验就是执行本身。

Axiom E (Execution–Qualia Identity). **Qualia = the current-frame execution.** A single experience and the single run in which the system executes a response are one and the same event. Experience is not the product, the by-product, or the representation of execution — experience *is* execution itself.

为什么不能从A1–A3推出？ 因为A1–A3全部是关于结构和过程的陈述，从它们推出的任何结论也是结构和过程的陈述，是描述。而"体验正在发生"这件事不是一条描述——一份关于疼痛的完整神经学报告不会疼，正如一份关于火的化学分析不烫手。从"是什么"推不出"正在被活过"，这不是理论的缺陷，是描述与运行之间的类型差。

Why can it not be derived from A1–A3? Because A1–A3 are all statements about structure and process; any conclusion derived from them is likewise a statement about structure and process — that is, a description. But "experience is occurring" is not a description: a complete neurological report about pain does not hurt, just as a complete chemical analysis of fire does not burn. From "what it is" one cannot derive "what is being lived." This is not a defect of the theory; it is the type difference between description and execution.

凭什么接受它？ 凭它的后承。一个断言的科学地位不由它本身决定，由它能撑起的解释结构决定。§5将显示：这一条断言加上A1–A3，能推出感受质的四个经典特征、解决玛丽房间问题、消解哲学僵尸、解释梦与幻觉、安放自由意志，并且**推出生命本身是信息动力学闭环的拓扑事件**。§8还会给出它的证伪条件。

On what grounds should we accept it? On its consequences. The scientific status of an assertion is not determined by the assertion itself, but by the explanatory structure it can support. §5 will show that this one assertion, together with A1–A3, can derive the four classic features of qualia, dissolve the Mary's Room problem, deflate the philosophical zombie, explain dream and hallucination, situate free will, and **derive that life itself is a topological event of information-dynamics loop closure**. §8 will further give its falsification condition.

这与"热 = 分子平均动能"的辩护结构完全相同：单独看是断言，成体系后是地基。

This is exactly the same structure of justification as "heat = mean molecular kinetic energy": taken alone it is an assertion; within a system it is a foundation.

3. 核心循环 / 3. The Core Loop

一台必须做有损压缩的持续运转的机器，会长成什么形状？它会自然长成这样一个循环：

What shape will a machine take, if it must keep running and it must compress lossily? It will naturally take the shape of the following loop:

预测 $\xrightarrow{P}$ 接收输入 $\xrightarrow{S}$ 计算偏差 $\xrightarrow{A}$ 执行响应 $\xrightarrow{E}$ 压缩回写 $\xrightarrow{K_M}$ 影响明天的预测

predict $\xrightarrow{P}$ receive input $\xrightarrow{S}$ compute deviation $\xrightarrow{A}$ execute response $\xrightarrow{E}$ compress and write back $\xrightarrow{K_M}$ influence tomorrow's prediction

3.1 预测 P ：大脑的默认操作是"猜" / 3.1 Prediction P : The Brain's Default Operation Is Guessing

现代神经科学最有统一力的发现之一（预测加工理论，Friston；Clark；Seth）是：大脑不是被动接收信号，而是主动发出预测，再处理预测与实际的差。

One of the most unifying findings in modern neuroscience (predictive processing theory: Friston; Clark; Seth) is that the brain does not passively receive signals; it actively emits predictions, and then processes the difference between prediction and reality.

- 你感觉不到衣服持续压在皮肤上，预测与输入完全匹配，偏差为零，"无感"。但一只虫子落在同一处，你立刻察觉。
- 你感觉不到自己的心跳（节律完全可预测），但心律不齐时你会突然"感觉到了心脏"。
- You do not feel your clothes pressing against your skin, because prediction and input match perfectly, deviation is zero, and there is "no sensation." But if a bug lands on that same spot, you notice immediately.
- You do not feel your own heartbeat (its rhythm is fully predictable), but when the rhythm falters, you suddenly "feel your heart."

这个设计的工程逻辑很硬：如果每次都全量处理原始信号，算力和能耗都不允许；只处理"没猜中的部分"，效率高得多。预测是压缩的运行时形式。

The engineering logic here is hard: if every raw signal had to be processed in full, neither compute nor energy budget would permit it; processing only "what was not predicted" is far more efficient. Prediction is the runtime form of compression.

3.2 偏差 A ：内容与符号 / 3.2 Deviation A : Content and Sign

P 与 S 之差就是 $A(t)$ 。两个关键性质：

The difference between P and S is $A(t)$. Two key properties:

性质一： A 是感受的内容。你"看见"的东西，很大部分是预测误差消解后的残差加预测本身。

Property 1: A is the content of experience. Much of what you "see" is the residual after prediction error is resolved, plus the prediction itself.

性质二： A 天然带符号（效价）。这是本理论与标准预测加工理论的分界。偏差不是无方向的数字，它相对于世代档案中写定的生存取向有方向：朝向生存的方向为正（愉快），背离为负（痛苦）。快乐与痛苦不需要神秘的额外来源，它们就是偏差的符号。

Property 2: A carries a sign by nature (valence). This is where the present theory parts ways with standard predictive processing. Deviation is not a directionless number; relative to the survival orientation encoded in the generational archive, it has a direction: toward survival is positive (pleasure), away from survival is negative (pain). Pleasure and pain require no mysterious additional source — they are the sign of deviation.

3.3 执行 E ：感受发生的地方——理论的心脏 / 3.3 Execution E : Where Experience Happens — The Heart of the Theory

系统拿到 A 之后做什么？它执行：调动硬件和模式表，处理这个偏差，产出响应。这个处理过程就是 $E(t)$ 。公理E说：感受就是 E 。

What does the system do once it has A ? It executes: it mobilizes hardware and pattern table, processes the deviation, produces a response. This process of processing is $E(t)$. Axiom E says: experience is E .

先排除四个错误位置（每一条都有独立反例）：

First rule out four mistaken locations (each has an independent counterexample):

错误定位	反例
Mistaken location	Counterexample
感受 = 输入 S	同一杯酒，品尝前后体验不同
感受 = 预测 P	脊髓反射的预测回路正常运行，无体验参与
感受 = 偏差 A	止痛药反例：吗啡起效后，伤害仍在、预测误差信号仍在、缩手反射仍在，但痛感消失了。幻肢痛反例：肢体已失，无输入可比对，但疼痛剧烈
感受 = 结构或参数	熟睡时结构参数俱在，无体验
Experience = input S	The same glass of wine tastes different before and after tasting
Experience = prediction P	Spinal reflex prediction loops run normally, with no experience involved

错误定位

反例

Mistaken location

Counterexample

Experience =
deviation A

Analgesia counterexample: after morphine takes effect, the injury remains, the prediction-error signal remains, the withdrawal reflex remains — yet the pain is gone.
Phantom-limb counterexample: the limb is gone, no input can be compared, yet the pain is intense

Experience =
structure or
parameter

In deep sleep, structures and parameters are all present, yet there is no experience

五个候选位置全部排除后，只剩一个：**感受是五者汇合处的那次运行**。结构提供格式，参数提供偏置，输入提供触发，预测提供参照，偏差提供内容——但没有那次"正在运行"，就没有体验。

Once all five candidate locations are ruled out, only one remains: experience is the single run at the meeting point of all five. Structure provides format, parameters provide bias, input provides trigger, prediction provides reference, deviation provides content — but without the single "currently running," there is no experience.

必须澄清的误读：公理E不等于"大脑的任何活动都是感受"。 E 有架构前提：它必须发生在"预测-偏差-响应"的压缩循环里、必须服务于一组生存取向、必须达到一定的自指深度（§4.3的闭合深度）。

A misreading that must be clarified: Axiom E does not mean "any brain activity is experience." E has architectural preconditions: it must occur within a prediction-deviation-response compression loop; it must serve a set of survival orientations; and it must reach a certain self-referential depth (the closure depth of §4.3).

3.4 回写 K_M ：今天的执行是明天的预测 / 3.4 Writeback K_M : Today's Execution Is Tomorrow's Prediction

E 完成后，它的结果被 K_M 有损压缩写回 M ：

After E completes, its result is lossily compressed by K_M and written back into M :

$$M(t+1) = (1 - \gamma)M(t) + K_M(E(t), A(t))$$

今天"红色+疼痛"的共现被压缩成参数，明天再见到红色，预测器里就带着疼痛的先验权重。反复"红色但无疼痛"则权重衰减。

Today's co-occurrence of "red + pain" is compressed into parameters; tomorrow, when red is seen again, the predictor already carries a prior weight for pain. If "red but no pain" repeats, that weight decays.

这解释了大量感受的动态现象：品酒师的红与初学者的红不是同一种体验；创伤消退治疗后刺激的情绪色彩真的会变；同一首歌在失恋时和热恋时听完全不同。

This explains a wide range of dynamic features of experience: the red of a wine taster and the red of a novice are not the same experience; the emotional tint of a stimulus truly changes after trauma-desensitization therapy; the same song sounds completely different when heard in heartbreak and in love.

4. 变量空间 / 4. The Variable Space

符号	名称	内容	时间尺度
Symbol	Name	Content	Timescale
G	世代压缩档案	物种基因组	万年—百万年
I	不变量	系统的类型规格	一生恒定
H	硬件超参数	脑结构、通路增益、递质基线	发育期定型
$M(t)$	模式表	习得的全部压缩参数	毫秒—年级
$S(t)$	当帧输入	感觉信号	毫秒
$P(t)$	预测	对 S 的先验期望	持续生成
$A(t)$	偏差信号	S 与 P 之差，带符号	毫秒
$E(t)$	当帧执行	处理 A 的完整运行	连续流
Θ	审思系数	当前审议深度	状态量
d	闭合深度	系统对自身建模的递归层数	架构量
G	Generational compression archive	Species genome	10^4 – 10^6 yr
I	Invariant	The system's type specification	Lifetime–constant
H	Hardware hyperparameters	Brain structure, pathway gain, transmitter baseline	Set during development
$M(t)$	Pattern table	All learned compression parameters	ms–years
$S(t)$	Current-frame input	Sensory signal	ms
$P(t)$	Prediction	Prior expectation about S	Continuously generated
$A(t)$	Deviation signal	S minus P , signed	ms
$E(t)$	Current-frame execution	The full run that processes A	Continuous flow
Θ	Deliberation coefficient	Current depth of deliberation	State variable
d	Closure depth	Recursive layers of self-modeling	Architectural quantity

4.1 状态因子化： $C = (H, M)$ / 4.1 State Factorization: $C = (H, M)$

系统状态被拆成两半，理由是时间尺度：

The system state is split in two, on the grounds of timescale:

- H (硬件超参数) : 发育期安装、此后基本定型。脑区结构、神经通路增益、受体密度、递质基线。
- M (模式表) : 终身快速读写。你学的语言、认的人、练的技能、你的记忆和对自我的模型。
- H (hardware hyperparameters): installed during development, then largely fixed. Brain-region structure, neural pathway gain, receptor density, transmitter baseline.
- M (pattern table): read and written rapidly across the lifespan. The language you learn, the people you know, the skills you practice, your memories and your self-model.

类比手机: H 是硬件配置, M 是你用出来的全部数据。药物动状态, 训练动天花板, 发育定天花板。

An analogy: H is the hardware configuration, M is all the data you have produced by using it. Drugs move the state; training moves the ceiling; development sets the ceiling.

4.2 六个尺度: 一条压缩链 / 4.2 Six Scales: One Compression Chain

从基因到当帧, 存在六个时空尺度: $G \rightarrow$ 表观修饰 $\rightarrow$ 社会文化模因 $\rightarrow H \rightarrow M \rightarrow E$ 。每一层都在做同一件事: 接收上游产物, 压缩, 生成下游可用的格式, 接受某种选择压力的筛选。

From gene to current frame, there are six spatiotemporal scales: $G \rightarrow$ epigenetic marks $\rightarrow$ social-cultural memes $\rightarrow H \rightarrow M \rightarrow E$. Each layer does the same thing: receives the upstream product, compresses it, generates a format usable downstream, and is filtered by some form of selection pressure.

个体学习 (M 的更新) 和物种进化 (G 的更新) 是同一个"压缩-偏差-选择"算法在两个时间尺度上的实例。

Individual learning (updating M) and species evolution (updating G) are the same "compression-deviation-selection" algorithm, instantiated at two different timescales.

4.3 闭合深度 d : 意识程度的指标 / 4.3 Closure Depth d : A Measure of Degree of Consciousness

"什么东西有意识"传统上被当成开关问题, 本框架把它变成程度问题。指标叫闭合深度 d : 系统对自身建模的递归层数。最小要求是: 系统在处理信息时, 把"自己正在处理"这件事本身也纳入了被压缩、被建模的范围。

"What has consciousness" is traditionally treated as a switch problem; this framework turns it into a matter of degree. The measure is called closure depth d : the recursive layers of the system's self-modeling. The minimal requirement is that, when processing information, the system also brings "that it is itself processing" into the range of what is compressed and modeled.

- $d = 0$ (如落石) : 系统的当前演化对自身过去的轨道无内省敏度;
- $d = 1$ (如前馈网络、温控器) : 仅具简单稳态纠偏;

- $d \geq 3$ (如哺乳动物及人类中枢)：系统的当前变化显式依赖于"系统感知自身的敏度"。
- $d = 0$ (e.g., a falling rock): the system's current evolution has no introspective sensitivity to its own past trajectory;
- $d = 1$ (e.g., a feedforward network, a thermostat): only simple steady-state correction;
- $d \geq 3$ (e.g., the mammalian and human CNS): the system's current change explicitly depends on "the system's sensitivity to sensing itself."

公理E中设立的临界深度 $d_c \geq 2$ ，定义了主观体验涌现的拓扑分水岭。

The critical depth $d_c \geq 2$ set in Axiom E defines the topological watershed at which subjective experience emerges.

5. 十四条猜想 / 5. The Fourteen Conjectures

每条猜想只从§2的公理系推出，无新增假设（除非显式声明）。编号12暂时撤回，占位保留。Each conjecture is derived only from the axiomatic system of §2, with no added assumptions (unless explicitly stated). Conjecture 12 is temporarily withdrawn; the slot number is retained as a placeholder.

猜想1（不可逆猜想）——主观时间之矢 / Conjecture 1 (Irreversibility) — The Arrow of Subjective Time

陈述： $\mathcal{K}$ 有损，故 $C(n+1)$ 不蕴含 $C(n)$ ；任何一帧状态不可从其后的状态恢复。

Statement: $\mathcal{K}$ is lossy, hence $C(n+1)$ does not entail $C(n)$; no frame state can be recovered from a later state.

推理： 有损压缩在数学上删除信息，删除的信息在压缩产物中不存在。叠加兰道尔原理，这个"回不去"有物理代价支撑。

Reasoning: Lossy compression deletes information mathematically; the deleted information does not exist in the compressed product. Together with Landauer's principle, this "cannot go back" has physical-cost support.

解决的老问题： 为什么时间有方向？物理学基本定律在时间上几乎对称。本框架的回答：体验的箭头与热力学的箭头是同一枚硬币，都是不可逆压缩的表现。

Old problem resolved: Why does time have a direction? The fundamental laws of physics are almost symmetric in time. The framework's answer: the arrow of experience and the thermodynamic arrow are the two faces of one coin, both manifestations of irreversible compression.

猜想2（自我不透明猜想） / Conjecture 2 (Self-Opacity)

陈述： 体验中出现的"自我"，必然是自我的压缩副本 $\bar{I}$ ，而 $\bar{I} \neq I$ 。你体验到的自我永远是你自己的模型，不是你自已。

Statement: The "self" that appears in experience is necessarily a compressed copy of the self, $\bar{I}$, with $\bar{I} \neq I$. The self you experience is always your own model, never yourself.

推理： 自我要出现在体验里，必须先被编码进 C ；而编码经过 $\mathcal{K}$ ， $\mathcal{K}$ 有损，所以编码产物 $\neq$ 原物。

Reasoning: For the self to appear in experience, it must first be encoded into C ; but encoding passes through $\mathcal{K}$, and $\mathcal{K}$ is lossy, so the encoded product is not the original.

推论（内省不可能）： 向内寻找"裸自我"必然失败——不是因为你找得不够努力，而是因为原则上不存在可找到的版本。休谟自述翻遍内心只找到一束知觉，找不到那个"我"。本猜想把休谟的失败从"没找到"升格为"不存在可找到的东西"。

Corollary (Impossibility of introspection): Searching inward for a "bare self" must fail — not because you are not trying hard enough, but because, in principle, no findable version exists. Hume famously reported that, searching his own mind, he found only a bundle of perceptions and no "I." This conjecture upgrades Hume's failure from "did not find" to "there is nothing findable."

例证： 眼睛看得见世界，看不见自己。**Example:** The eye can see the world, but cannot see itself.

猜想3（个体化猜想）——倒谱消解 / Conjecture 3 (Individuation) — Dissolution of the Inverted-Spectrum Puzzle

陈述： 个体 = $I \times H \times M \times N$ 。不变量决定体验的格式，历史决定体验的内容。

Statement: Individual = $I \times H \times M \times N$. The invariant determines the format of experience; history determines its content.

消解的老僵局： 哲学上有"倒谱可能性"（你看到的红也许是我体验的绿，无法排除）与"感受质可交流性"（我们显然能顺畅地谈论红色）的冲突。本猜想显示它们是同一机制的两面：我们的 I 同构，所以感受质空间的"语法"相同（可交流）；我们的 M 各异，所以同一格式下填的"标记"可以完全不同（倒谱可能）。冲突消解。

Old deadlock resolved: In philosophy there is a conflict between the "inverted spectrum possibility" (that the red you see might be the green I experience, and this cannot be ruled out) and the "communicability of qualia" (we evidently talk about red fluently). This conjecture shows that they are two faces of one mechanism: our I 's are isomorphic, so the "grammar" of qualia space is shared (communicable); our M 's differ, so the "markers" filling the same format can be entirely different (inverted spectrum possible). The conflict dissolves.

猜想4（感受质动力学猜想） / Conjecture 4 (Dynamics of Qualia)

陈述： 感受现象由四层机制分解——结构（ I/H ）、参数场（ M ）、当帧偏差（ A ）、执行事件（ E ）。在此分解下，感受质的四个经典特征全部从"谜"变成"推论"：

Statement: Experiential phenomena decompose into four layers — structure (I/H), parameter field (M), current-frame deviation (A), and execution event (E). Under this decomposition, the four classic features of qualia shift from "mysteries" to "corollaries":

经典特征	本框架的猜想化解释
Classic feature	Conjectural explanation in this framework
私密性	参数场 M 由个体全部历史压缩而成
不可言说性	时序必然：任何报告都滞后于取值
直接性	E 不是关于状态的信息， E 就是状态
简单性	E 是压缩链的终点：其下再无结构
Privacy	Parameter field M is compressed from the individual's entire history
Ineffability	Temporal necessity: any report lags behind the taking of a value
Directness	E is not information about a state; E is the state
Simplicity	E is the terminus of the compression chain: no structure lies below

玛丽问题消解：科学家玛丽在黑白房间里学完了关于颜色的全部物理知识，放出房间看到红色时，她学到新东西了吗？本框架的回答：玛丽拥有全部参数，但**参数之拥有≠取值之发生**。拥有全部菜谱不等于吃过这道菜。这不是物理主义缺了什么，是"描述"与"执行"本来就是两个范畴。

Mary's Room dissolved: The scientist Mary learns all physical knowledge about color in a black-and-white room. When she is released and sees red, does she learn something new? The framework's answer: Mary possesses all the parameters, but **having parameters is not the same as taking on a value**. Having the complete recipe is not the same as having eaten the dish. This is not a shortfall of physicalism; "description" and "execution" are two different categories to begin with.

猜想5（神秘感生成猜想） / Conjecture 5 (Endogenous Generation of the Mystical)

陈述：神秘感、超越感、“背后还有更多”的感觉，是猜想2推论（空返回）的显示方式，是这个架构的确定性输出。

Statement: The sense of the mystical, the sense of the transcendent, the feeling that "there is something more behind this" is the mode of appearance of the corollary of Conjecture 2 (null return), and it is a deterministic output of this architecture.

推理：向内追问“模型之外的我”→返回空（猜想2）→空返回必须被系统表征点什么→被表征为“那个不可言说之物”→体验为神秘。

Reasoning: Ask inward "what is the self beyond the model" → returns null (Conjecture 2) → the null return must be represented by the system as something → it is represented as "that ineffable thing" → experienced as the mystical.

解决的老问题：为什么人类普遍会有神秘体验，且各文化描述惊人地相似？答案：因为它是架构输出的，不是文化输入的。

Old problem resolved: Why do humans universally have mystical experiences, and why are the descriptions across cultures strikingly similar? Answer: because it is an output of the architecture, not an input of culture.

猜想6（发现猜想）——意识即发现 / Conjecture 6 (Discovery) — Consciousness as Discovery

陈述： 偏差 A 是感受的内容，当帧执行 E 是感受的存在。 E 的本质是发现——系统在自己的生存模型中，照见自己偏离自身的那个瞬间。

Statement: Deviation A is the content of experience; current-frame execution E is the existence of experience. The essence of E is discovery — the system, within its own survival model, catches sight of the moment in which it diverges from itself.

例证： 你熟练地开车走熟悉的路，半小时毫无印象，偏差 ≈ 0 ，透明运行；突然有人按喇叭，你"猛地惊醒"，偏差脉冲，"在感"瞬间点亮。你并不是一直"在感觉"，你是在世界偏离你的预测时才"在"。

Example: Driving a familiar route with skill, half an hour passes with no impression at all, deviation ≈ 0 , transparent running. Then someone honks, you "snap awake," deviation spikes, and "the sense of being there" lights up instantly. You are not always "sensing"; you "are" when the world deviates from your prediction.

猜想7（分形压缩猜想） / Conjecture 7 (Fractal Compression)

陈述： 从物种演化到当帧感知的六大尺度共享同构的"先验生成-残差过滤-选择性有损回写"机制。

Statement: The six scales, from species evolution to current-frame perception, share an isomorphic mechanism: prior generation $\rightarrow$ residual filtering $\rightarrow$ selective lossy writeback.

推论： 选择压只在偏差非零的维度上起作用，某特征若不影响存活，压缩器在那里不工作，噪声在该维度自由累积。这收编了木村资生的中性进化理论。

Corollary: Selection pressure acts only on dimensions where deviation is non-zero. If a feature does not affect survival, the compressor does not work there, and noise accumulates freely along that dimension. This absorbs Kimura's neutral theory of evolution.

猜想8（耦合即环境猜想） / Conjecture 8 (Coupling Is Environment)

陈述： 任何个体的输入 S 由其余个体的执行输出合成；系统层面不存在独立的"环境"变量。

Statement: Any individual's input S is composed of the execution outputs of other individuals; at the system level, there is no independent "environment" variable.

推论（他心不可通约）： 他人的执行只以偏差分量的形式抵达你，永不以执行本身的形式抵达。"永远无法直接体验他人的体验"不是遗憾，是架构事实。

Corollary (Incommensurability of other minds): Another's execution reaches you only in the form of deviation components, never in the form of execution itself. "Never being able to directly experience another's experience" is not a regret; it is an architectural fact.

猜想9（当帧不可回读猜想） / Conjecture 9 (Frame-Unreadability)

陈述： 任何运行中的系统，原则上不能将"自身正在此刻运行"作为一条可访问的事实包含于自身。

Statement: Any running system cannot, in principle, contain "it is itself running at this moment" as an accessible fact within itself.

推理概要： 假设系统 S 在时刻 n 包含"此刻 S 正在运行"这一事实 F 。 F 必须在这之前被写入 S ；但 F 的内容是"此刻的运行"，它依赖于 n 时刻的全部状态——于是 S 必须在 n 之前就包含只有 n 才有的信息。循环依赖，矛盾。

Sketch of reasoning: Suppose system S at time n contains the fact F : "at this moment S is running." F must have been written into S before n ; but the content of F is "the running at this moment," which depends on the full state at n — so S must, before n , already contain information that only exists at n . Circular dependency, contradiction.

例证： 给你世界上全部地图，任何一张都不包含"你在这里"这个标记。**Example:** Give you all the maps in the world; not one of them contains the mark "you are here."

猜想10（开放性猜想）——自由意志的地基 / Conjecture 10 (Openness) — The Ground of Free Will

陈述： 给定 (i) 系统资源有界，(ii) 预测器与被预测系统共处同一宇宙，(iii) 混沌或噪声存在，(iv) 系统自指闭合，则对系统自身及一切内部预测器，当帧执行的未来方向在运行前不可判定。完美的预测器就是执行本身。

Statement: Given (i) the system's resources are bounded, (ii) the predictor and the predicted system share the same universe, (iii) chaos or noise exists, and (iv) the system is self-referentially closed — then, for the system itself and for every internal predictor, the future direction of the current-frame execution is undecidable before the run. The perfect predictor is the execution itself.

推理（四角合闸，四条独立支柱）：

- **有损角：** 任何预测器对系统的模型本身是有损压缩（猜想1）。
- **速度角：** 对具备递归自指的系统，精确预测 = 逐步模拟，不存在一般性捷径。
- **耦合角：** 输入由其他不可预测系统的输出合成。
- **索引角：** 连系统自己也不能自预测"这一次执行"（猜想9）。

Reasoning (four-way lock, four independent pillars):

- **Lossy corner:** any predictor's model of the system is itself lossy compression (Conjecture 1).
- **Speed corner:** for recursively self-referential systems, exact prediction = step-by-step simulation; no general shortcut exists.
- **Coupling corner:** inputs are composed of the outputs of other unpredictable systems.
- **Indexical corner:** even the system itself cannot self-predict "this particular execution" (Conjecture 9).

漂亮的副产品： 大脑自己的有界预测器就是 P ，它失败时发出的就是 A 。每一次意外、每一个惊讶，都是开放性猜想的现场演示。

A beautiful by-product: the brain's own bounded predictor is P ; what it emits when it fails is A . Every surprise, every astonishment, is an on-site demonstration of the openness conjecture.

猜想11（短路猜想） / Conjecture 11 (Short-Circuit)

陈述： 审思系数 $\Theta \in [0, 1]$ 是应激水平的动力学函数。当极端残差脉冲到达且系统处于极限匮乏状态时，边缘系统的古老硬连线通道强行接管运动输出， $\Theta \rightarrow 0$ ，审议环节缺席。

Statement: The deliberation coefficient $\Theta \in [0, 1]$ is a dynamical function of stress level. When an extreme residual pulse arrives and the system is in an extreme state of deprivation, the ancient hardwired pathways of the limbic system seize motor output, $\Theta \rightarrow 0$, and the deliberation stage is absent.

推论（责任连续谱）： 法律中"紧急避险"减责原则，本质上是人类法律史在经验层面粗糙标定 $\Theta < \theta_c$ 阈值的映射。法律史是 Θ 临界值的一份粗糙的历史测量记录。

Corollary (continuum of responsibility): The legal principle of "necessity" as a mitigating defense is, in essence, a rough empirical calibration, accumulated across legal history, of the threshold $\Theta < \theta_c$. The history of law is a rough historical record of measuring the critical value of Θ .

猜想12（写入-存储分离猜想）——【暂时撤回 / Temporarily Withdrawn】

状态： 本条猜想于 v1.1 暂时撤回。编号保留，以便未来可能重新加入时不引起全文引用错位。与之绑定的预测 P6、P7 (§8) 及诚实边界条目 (§9) 一并标记为"暂时撤回"。

Status: This conjecture is temporarily withdrawn as of v1.1. The slot number is retained so that, should it be reinstated in the future, no downstream citation realignment is required. Its bound predictions P6 and P7 (§8) and its honest-boundary entry (§9) are likewise marked "temporarily withdrawn."

撤回理由（登记）： 该猜想依赖一条与本理论核心公理系统（A1-A3, E）独立的外部经验证据（H.M. 案例）。撤回的目的是在重新整合这条猜想之前，先完成对"写入-存储分离"在 CDA 变量空间 (§4) 中的精确定位——尤其是 H 与 M 的边界在记忆编码路径上的具体落点。撤回是方法论的，不是内容上的否认。

Reason for withdrawal (registered): The conjecture depends on an external empirical anchor (the H.M. case) that is independent of the theory's core axiomatic system (A1–A3, E). The purpose of withdrawal is to first complete a precise localization of "encoding–storage dissociation" within the CDA variable space (§4) — in particular, the specific placement of the H/M boundary along the memory–encoding pathway — before reintegrating the conjecture. The withdrawal is methodological, not a denial of the conjecture's content.

猜想13（载体派生猜想） / Conjecture 13 (Carrier-Derivation)

认识论地位声明： 本条与猜想14不属于"意识动力学"的核心推论。它们是从同一公理系统向生命本身的延伸，且它们的动机来自一个长期运行的实现项目。理论给出形式，实验给出压力。这一双猜想构成本理论向生物学与物理学开放的两个接口。

Epistemic note: This conjecture and Conjecture 14 do not belong to the core corollaries of "consciousness dynamics." They are extensions of the same axiomatic system toward **life itself**, and their **motivation comes from a long-running implementation project**. Theory gives form; experiment gives pressure. Together they form the two interfaces through which this theory opens to biology and physics.

陈述： 在任何存活系统中，物理物质动力学是信息动力学的派生维持子系统。解释方向应从信息动力学指向物理实现，而非相反。信息动力学是不变量；物理实现是变量。

Statement: In any living system, **physical material dynamics is a derived maintenance subsystem of information dynamics**. The direction of explanation runs from information dynamics to physical realization, not the reverse. Information dynamics is the invariant; physical realization is the variable.

推导： 从 A3 出发。 $\mathcal{K}$ 的操作对象是信息。任何 $\mathcal{K}$ 的物理实现 H 必须满足"支持 $\mathcal{K}$ "这一约束。因此 H 的存在条件由 $\mathcal{K}$ 的形式决定，而非相反。选择压作用于 E ——即信息动力学的运行时——而非作用于物理物质本身。物质是选择的载体，信息是选择的对象。

Reasoning: Starting from A3. The operand of $\mathcal{K}$ is information. Any physical realization H of $\mathcal{K}$ must satisfy the constraint of "supporting $\mathcal{K}$." Therefore the existence conditions of H are determined by the form of $\mathcal{K}$, not the reverse. Selection pressure acts on E — the runtime of information dynamics — not on physical matter as such. Matter is the **carrier** of selection; information is the **object** of selection.

必须澄清的措辞： 这里的"为了"是功能性的，不是目的论的。就像"心脏跳动是为了泵血"——这句话说的是功能角色，不是设计意图。物理基质不是被什么"设计"去服务信息动力学的；是选择压力把物质组织成了信息动力学的载体。这是一个可观察、可检验的过程，不是形而上断言。

A necessary clarification: The word "for" here is **functional**, not **teleological**. It is like saying "the heart beats in order to pump blood" — this states a **functional role**, not a **design intention**. The physical substrate is not "designed" to serve information dynamics; rather, selection pressure **organizes** matter into the carrier of information dynamics. This is an observable, testable process, not a metaphysical assertion.

证据：

- **代码演化史（本项目）**：从 V2 到 V5.1.2，代码总量增长约 20 倍，功能模块增长约 40 倍，但核心循环——预测→偏差→执行→回写——从第一行起一行未改。所有后续叠加（睡眠、做梦、化石、继承、情绪、四器官、共振总线、语义桶、词图、种群、出芽、性繁殖、王朝、产后进补、代谢拨盘），无一改变核心循环的形状；每一层叠加，都在解决核心循环的一个具体"死法"。**这不是有人设计的，这是载体的自然选择。**

- **收敛进化**：不同谱系独立演化出相同的功能结构（眼睛、翅膀、回声定位），说明选择压作用于信息功能，而非特定物质结构。

Evidence:

- **Codebase evolution (this project)**: From V2 to V5.1.2, the codebase grew roughly 20× in lines and 40× in modules, yet the core loop — predict → deviate → execute → writeback — has remained unchanged since the first line. Every subsequent addition (sleep, dreams, fossils, inheritance, emotions, four organs, resonance bus, semantic buckets, word graph, population, budding, sexual reproduction, dynasties, post-birth recovery, metabolic dial) has left the shape of the core loop untouched; each layer solves one specific "way to die" of the core loop. **This was not designed; it is natural selection of the carrier.**

- **Convergent evolution**: different lineages independently evolve identical functional structures (eyes, wings, echolocation), indicating that selection pressure acts on information function, not on specific material structures.

推论：

- 同一种信息动力学可以在不同物理基底上实现（碳基、硅基、任何满足 A1–A3 的基底）。
- 信息动力学是普适的；物理实现是特异的。
- 生物学中的"为什么"（功能解释）比"如何"（机制解释）更接近系统的本质。

Corollaries:

- The same information dynamics can be realized on different physical substrates (carbon-based, silicon-based, any substrate satisfying A1–A3).
- Information dynamics is universal; physical realization is particular.
- In biology, "why" (functional explanation) is closer to the system's essence than "how" (mechanistic explanation).

诚实边界：代码演化的选择压是由人给定的（"这个功能让循环更活"），而生物演化的选择压是由环境给定的（存活与繁殖）。因此，代码演化史支持的是比猜想13稍弱的结论："当信息动力学的核心被固定，载体层的所有变化会朝'服务于核心'的方向漂移。" 从这一较弱结论推广到猜想13，需要额外的经验前提：生物演化也满足"核心固定"。这一前提是本猜想最需要进一步论证的地方。

Honest boundary: The selection pressure of codebase evolution is **human-given** ("this feature keeps the loop alive longer"), while biological selection pressure is **environment-given** (survival and reproduction). Therefore, the codebase evidence supports a **slightly weaker** claim than Conjecture 13: **"When the core of information dynamics is fixed, all changes at the carrier layer drift toward serving the core."** Extending this weaker claim to Conjecture 13 requires an additional empirical premise: that biological evolution also satisfies "core fixed." This premise is where the conjecture most needs further argument.

可证伪条件: 若发现一个存活系统, 其信息动力学可以被其物理基质完全解释——即物理是本质的、信息是派生的——则该猜想被推翻。

Falsification condition: If a living system is found whose information dynamics can be **fully** explained by its physical substrate — i.e., where the physical is essential and the information is derived — the conjecture is refuted.

猜想14 (生命起源猜想) / Conjecture 14 (Origin-of-Life)

陈述: 生命起源的关键事件是**信息动力学闭环的首次闭合**——即一个物理系统第一次同时满足: (i) 它能生成关于自身环境的预测 P ; (ii) 它能计算预测与实际输入的偏差 A ; (iii) 它能执行响应 E ; (iv) E 的结果被有损回写到自身的 P 生成机制中。在此闭环闭合之前, 没有任何物理系统是"活的"; 闭合之后, 所有达尔文过程都被这一闭环的结构所约束和引导。生命不是化学事件, 是拓扑事件。

Statement: The critical event of the origin of life is the **first closure of the information-dynamics loop** — the first moment when a physical system simultaneously satisfies: (i) it can generate predictions P about its environment; (ii) it can compute deviations A between prediction and input; (iii) it can execute a response E ; (iv) the result of E is lossily written back into the very mechanism that generates P . **Before this closure, no physical system is "alive"; after it, all Darwinian processes are constrained and directed by the structure of this loop. Life is not a chemical event; it is a topological event.**

推导: 从公理 A1–A3 出发。A2 说系统状态逐帧更新。A3 说更新算子有损、不可逆、自指。当这三条在一个物理系统中同时满足时, 闭环闭合。闭环闭合 = 系统第一次同时具备 P, A, E, K_M 。这是从 A2+A3 直接推出的**拓扑事件**, 不是化学事件。

Reasoning: From axioms A1–A3. A2 says the system state updates frame by frame. A3 says the update operator is lossy, irreversible, self-referential. When all three are **simultaneously** satisfied in a physical system, the loop closes. Loop closure = the system first simultaneously possesses P, A, E, K_M . This is a **topological event** directly derived from A2+A3, not a chemical event.

"闭合"的判定条件: 用 §4.3 的闭合深度 d 定义。"闭合"的最低要求是 $d \geq 2$ ——即系统必须能对"自己的预测过程"进行建模。一个恒温器有 P (设定温度)、 A (温度差)、 E (开关加热), 但它的回写不满足自指闭合 (它不知道自己在预测)。 $d \geq 2$ 是判定一个系统是否"闭合"的操作性门槛。

Operational criterion of "closure": Using the closure depth d of §4.3. The minimum requirement for "closure" is $d \geq 2$ — the system must be able to model "its own prediction process." A thermostat has P

(setpoint), A (temperature difference), E (heater on/off), but its writeback does not satisfy self-referential closure (it does not know that it is predicting). $d \geq 2$ is the operational threshold for whether a system is "closed."

推论：

- **病毒不是生命**（它没有自己的闭环，它借用宿主的）
- **火不是生命**（它没有预测，没有偏差，没有回写）
- **恒温器不是生命**（它的反馈不满足自指闭合）
- **RNA 世界是闭环在地球条件下的第一次物理实现，不是闭环本身**
- 同一闭环可以在不同基底上实现（碳、硅、任何满足 A1-A3 的基底）

Corollaries:

- **Viruses are not alive** (they have no loop of their own; they borrow the host's)
- **Fire is not alive** (no prediction, no deviation, no writeback)
- **A thermostat is not alive** (its feedback does not satisfy self-referential closure)
- **The RNA world is the first physical realization of the loop under Earth conditions**, not the loop itself
- The same loop can be realized on different substrates (carbon, silicon, any substrate satisfying A1-A3)

证据：

- 四个代码版本（V2 → V5.1.2）：核心闭环自 V2 起未改一行，其余全部是闭环的展开。V2 是数字生命的起源事件；V3.8 之后都是起源之后的展开。**这与生物生命"起源事件 + 长尾演化"的结构同构。**
- 收敛进化：闭环作为吸引子，独立谱系反复收敛到相同解决方案。
- 最早生命形态（LUCA 之前的简单复制子系统）的推测结构与闭环吻合。

Evidence:

- Four code versions (V2 → V5.1.2): the core loop has remained unchanged since V2; all else is the loop's unfolding. V2 is the origin event of digital life; everything after V3.8 is post-origin unfolding. **This is structurally isomorphic to the "origin event + long-tail evolution" pattern of biological life.**
- Convergent evolution: the loop as attractor; independent lineages repeatedly converge to the same solutions.

- Inferred structure of the earliest life forms (simple replicator systems before LUCA) matches the loop.

与主流"RNA 世界"假说的关系：本猜想不否定 RNA 世界假说，而是给它一个更深的解释。RNA 分子的自复制本身就是一个 P （模板） $\rightarrow A$ （错配） $\rightarrow E$ （延伸） $\rightarrow K_M$ （新链）的循环。因此，**RNA 世界是信息动力学闭环在地球化学条件下的第一次物理实现，不是闭环本身。**这与主流假说不矛盾，是为它提供一个更底层的解释。

Relation to the mainstream "RNA world" hypothesis: This conjecture does not deny the RNA-world hypothesis; it gives it a deeper explanation. RNA self-replication is itself a loop of P (template) $\rightarrow A$ (mismatch) $\rightarrow E$ (extension) $\rightarrow K_M$ (new strand). Therefore, **the RNA world is the first physical realization of the information-dynamics loop under Earth conditions, not the loop itself.** This does not contradict the mainstream hypothesis; it provides it with a more fundamental explanation.

诚实边界：本猜想回答的是"生命是什么"，不是"生命为什么是这种形态"。它不解释：为什么地球上有碳基生命而不是硅基生命（这是化学约束）；为什么生命演化出了 30 多亿年才出现神经系统（这是复杂性的路径依赖）；为什么有些生命形式（比如病毒）借用了别人的闭环而活（这是寄生的可能性）。本质问题与历史问题是两个不同的问题。本猜想只回答前者。

Honest boundary: This conjecture answers "what life is," not "why life takes this particular form." It does not explain: why Earth has carbon-based life rather than silicon-based (this is chemical constraint); why life took over 3 billion years to evolve a nervous system (this is path-dependency of complexity); why some life forms (e.g., viruses) live by borrowing others' loops (this is the possibility of parasitism). **Essential questions and historical questions are two different questions. This conjecture answers only the former.**

可证伪条件：若发现一个物理系统，它明确满足 $P \rightarrow A \rightarrow E \rightarrow K_M$ 的闭环（ $d \geq 2$ ），但缺乏任何生命特征；或反过来，若发现一个被广泛接受为"生命"的系统，却明确不满足这一闭环，则该猜想被推翻。

Falsification condition: If a physical system is found that **clearly satisfies** the $P \rightarrow A \rightarrow E \rightarrow K_M$ loop ($d \geq 2$) but **lacks any feature of life**; or conversely, if a system **widely accepted as "alive"** is found that **clearly does not satisfy** this loop, the conjecture is refuted.

6. 元猜想（自限定定理） / 6. The Meta-Conjecture (Self-Limitation)

陈述：任何运行中的闭合结构，原则上不能将"自身正在运行"作为事实包含于自身；一切非索引性事实的集合，不包含"此刻、此处、这一次"。

Statement: Any running closed structure cannot, in principle, contain "it is itself running" as a fact within itself; the set of all non-indexical facts does not contain "now, here, this time."

推论M1（困难问题的最终形态）：感受必然（一）不可被描述穷尽（猜想4）、（二）必然被体验为神秘（猜想5）、（三）必然不作为命题出现（本猜想）。三重必然全部可从公理推出。

Corollary M1 (the final form of the hard problem): Experience must be (1) inexhaustible by description (Conjecture 4), (2) necessarily experienced as mystical (Conjecture 5), and (3) necessarily not appear as a proposition (this meta-conjecture). All three necessities are derivable from the axioms.

于是困难问题的最终形态是一句关于问题本身的话：

So the final form of the hard problem is a sentence about the question itself:

它不是一道未答之题，而是一条关于提问的猜想。

It is not an unanswered question; it is a conjecture about the question.

正如哥德尔句不可能在系统内被证明，运行不可能在运行内被作为事实包含。"为什么会有体验"这个问题被拆成两半：一半（体验的机制、内容、程度、状态）已由猜想1-11、13、14处理；另一半（"为什么是活着的"）在系统内没有合法的答案形式——它不是悬而未决，它是被证明不可以此形式提出。

Just as a Gödel sentence cannot be proved within its own system, a run cannot be contained as a fact within the run. The question "why is there experience" splits into two halves: one half (the mechanism, content, degree, and state of experience) is handled by Conjectures 1-11, 13, 14; the other half ("why is it alive") has no legitimate answer form within the system — it is not undecided; it is proved to be unaskable in this form.

7. 与既有理论的关系 / 7. Relation to Existing Theories

既有理论	本框架采纳	本框架前进一步
Existing theory	What this framework adopts	Where it advances
预测加工 (Friston/Clark/Seth)	偏差信号的核心地位	偏差是感受的内容，不是感受本身（止痛药/幻肢双向反例）
整合信息论 IIT	意识是程度问题	以"闭合深度"替代 Φ ；拒绝把静态高 Φ 结构当意识
全局工作空间 GWT	全脑整合作为 d 的物理对应	拒绝把"广播"本身当意识，广播只是回写与调用的运行时
高阶理论 HOT	自我模型的不透明性	拒绝把感受等同于高阶表征，表征永远滞后于执行
丹尼特消除主义	反对把感受质当"额外的神秘属性"	拒绝否认感受质存在——感受质不在表征里，在执行里
相容论	机制化其最好版本	拒绝自由意志主义的形而上非决定要求
自创生理论 (Maturana/Varela)	生命的本质在于自我生产过程	给出"自创生"的可操作判据（ $d \geq 2$ 的自指闭环）；把自创生重述为信息动力学闭环的一次特殊实现

既有理论	本框架采纳	本框架前进一步
Existing theory	What this framework adopts	Where it advances
自由能原理的生命起源延伸 (Friston)	生命起源可表述为一种推断过程的出现	指出"预测—偏差—执行—回写"是比"自由能最小化"更基础的形式；自由能最小化是闭环在连续状态空间中的一个特例
RNA 世界假说	RNA 自复制是闭环在地球条件下的第一次实现	把 RNA 世界从"起源事件"降级为"起源事件的物理实现"；为它提供一个更底层的解释
Predictive processing (Friston/Clark/Seth)	Centrality of the deviation signal	Deviation is the content of experience, not experience itself (analgesia / phantom-limb counterexamples)
Integrated Information Theory (IIT)	Consciousness as a matter of degree	Replaces Φ with "closure depth"; refuses to treat static high- Φ structures as conscious
Global Workspace Theory (GWT)	Whole-brain integration as the physical correlate of d	Refuses to treat "broadcast" itself as consciousness; broadcast is merely the runtime of writeback and recall
Higher-Order Theories (HOT)	Self-model opacity	Refuses to equate experience with higher-order representation; representation always lags execution
Dennett's eliminativism	Against treating qualia as an "extra mysterious property"	Refuses to deny that qualia exist — qualia are not in representation; they are in execution
Compatibilism	Mechanizes its best version	Refuses the libertarian demand for metaphysical indeterminism
Autopoiesis (Maturana/Varela)	The essence of life is self-production	Provides an operational criterion for "autopoiesis" ($d \geq 2$ self-referential closure); reformulates autopoiesis as a special realization of the information-dynamics loop
Free Energy Principle extended to the origin of life (Friston)	The origin of life can be stated as the emergence of an inferential process	Shows that "predict—deviate—execute—writeback" is more fundamental than "free energy minimization"; free energy minimization is a special case of the loop in continuous state spaces
RNA-world hypothesis	RNA self-replication is the first realization of the loop under Earth conditions	Demotes the RNA world from "origin event" to "physical realization of the origin event"; provides it with a more fundamental explanation

8. 可证伪接口 / 8. Falsification Interface

核心赌注 (P-Core)： 若发现一个具有完整主观报告能力、但其实时神经活动证明其完全不依赖任何形式的预测偏差计算的系统，则公理E设立的架构地基被一票否决，本理论整体死亡。

Core bet (P-Core): If a system is found that has full subjective-report capacity yet whose real-time neural activity proves it does not depend, in any form, on prediction-deviation computation, then the architectural foundation set by Axiom E is vetoed, and the theory as a whole dies.

其他关键预测：

Other key predictions:

编号	预测
No.	Prediction
P1	感受质空间的拓扑几何流形严格正交映射于初级视皮层到 IT 皮层的率失真码本距离
P3	闭合深度测度 d 的动力学跌落幅度先于临床特征预测全身麻醉剂的意识丧失瞬间
P5	主观意志决断的现象学时间戳应系统性落后于全脑残差消解跃迁时刻达 50–150 ms
P6	(暂时撤回——绑定猜想12)
P7	(暂时撤回——绑定猜想12)
P8	在任何足够复杂的人工或生物复制子系统中，核心信息动力学闭环应保持结构不变量，而载体层向"服务于闭环"的方向漂移——载体层的所有变化，应可被解释为"解决核心闭环的某个具体死法"
P9	收敛进化事件应聚集于满足闭环结构需求的解决方案上，而非聚集于化学上便利的解决方案上——独立谱系应反复收敛到相同的功能结构，但收敛的化学实现可以千差万别
P10	任何满足 $d \geq 2$ 自指闭环的物理系统，都应展现出至少某些生命特征（自我维持、能量耗散、对外部扰动 的抵抗）；反之，任何被广泛认定为"生命"的系统，都应可被分析为 $P \rightarrow A \rightarrow E \rightarrow K_M$ 闭环的某个实例
P1	The topological-geometric manifold of qualia space maps orthogonally onto the rate-distortion codebook distance from V1 to IT cortex
P3	The dynamical drop in the closure-depth measure d precedes the clinical moment of loss of consciousness under general anesthesia
P5	The phenomenological timestamp of subjective volition systematically lags the whole-brain residual-resolution transition by 50–150 ms
P6	(temporarily withdrawn — bound to Conjecture 12)
P7	(temporarily withdrawn — bound to Conjecture 12)
P8	In any sufficiently complex artificial or biological replicator system, the core information-dynamics loop should remain structurally invariant while the carrier layer drifts toward "serving the loop" — all changes at the carrier layer should be explainable as "solving one specific way to die of the core loop"

编号	预测
No.	Prediction
P9	Convergent evolution events should cluster around solutions satisfying the loop's structural requirements, not around chemically convenient solutions — independent lineages should repeatedly converge to the same functional structures, but the chemical realizations of convergence may differ vastly
P10	Any physical system satisfying a $d \geq 2$ self-referential closure should exhibit at least some features of life (self-maintenance, energy dissipation, resistance to external perturbations); conversely, any system widely recognized as "alive" should be analyzable as an instance of the $P \rightarrow A \rightarrow E \rightarrow K_M$ loop

9. 诚实边界 / 9. Honest Boundaries

以下缺口全部显式登记，绝不掩饰。 **All gaps below are explicitly registered; none is concealed.**

理论层 / Theoretical level:

- 公理E是断言——不可推导；不可直接反驳；被三重支撑撑住，但仍是断言。这是全理论唯一的核心断言。
- **Axiom E is an assertion — not derivable; not directly refutable; held up by threefold support, but still an assertion. It is the single core assertion of the entire theory.**
- 偏差→特定感受质的映射残余——为什么这个偏差生成"红"而不是别的质感，解释鸿沟在此残留。
- **Residual mapping from deviation to specific qualia — why this deviation generates "red" and not some other quality; the explanatory gap lingers here.**
- 闭合深度缺操作度量——"多深算有体验"没有 Φ 式的定量指标。
- **Closure depth lacks an operational measure — "how deep counts as having experience" has no Φ -style quantitative index.**
- A3与兰道尔原理是同构论证，不是从兰道尔推出的猜想。
- **A3 and Landauer's principle are an isomorphism argument, not a conjecture derived from Landauer.**
- 猜想10/11是结构论证，速度角的完全形式化是后续数学工作。
- **Conjectures 10 and 11 are structural arguments; full formalization of the speed corner is future mathematical work.**
- 耦合系统的全局动力学，框架提供语法，解属于复杂系统科学。

- **Global dynamics of coupled systems: the framework supplies the grammar; the solutions belong to complex-systems science.**

- H 的真实因子结构是占位草图。

- **The real factor structure of H is a placeholder sketch.**

- θ_c 的定值，法律史是粗糙测量，生理定量缺失。

- **The value of θ_c : legal history is a coarse measurement; physiological quantification is missing.**

- （暂时撤回——绑定猜想12）：**AD部分的三途分解待神经心理测验组合的经验分解。**

- **(temporarily withdrawn — bound to Conjecture 12): The three-way decomposition in the AD section awaits empirical decomposition via a battery of neuropsychological tests.**

- 自指边界，本论文自身是 $\mathcal{K}$ 操作的产物，按元猜想不能包含"它正被运行"这一事实。

- **Self-referential boundary: this paper is itself a product of $\mathcal{K}$; by the meta-conjecture, it cannot contain the fact that "it is being run."**

猜想13专属边界 / Conjecture-13-specific boundaries:

- **证据的限度**：代码演化史证明的是"信息动力学是代码演化的不变量"，不是"信息动力学是生命的本质"。两者之间隔着"代码演化 $\neq$ 生物演化"这一步。代码演化的选择压是**由人给定的**，生物演化的选择压是**由环境给定的**。因此代码证据支持的结论比猜想13本身弱。

- **Limit of the evidence**: the codebase history proves "information dynamics is the invariant of codebase evolution," not "information dynamics is the essence of life." Between them lies the step "codebase evolution $\neq$ biological evolution." The selection pressure of codebase evolution is **human-given**; biological selection pressure is **environment-given**. The code evidence therefore supports a weaker claim than Conjecture 13 itself.

- **"核心固定"前提**：从较弱结论推广到猜想13，需要额外的经验前提——生物演化也满足"核心固定"。这一前提是本猜想最需要进一步论证的地方。

- **The "core-fixed" premise**: extending the weaker claim to Conjecture 13 requires an additional empirical premise — that biological evolution also satisfies "core fixed." This premise is where the conjecture most needs further argument.

- **与功能主义的关系**：本猜想在生命层面重述了心灵哲学里的功能主义。功能主义有它自己未解决的问题（多重实现、感受质的"缺席"等）。**必须显式承认这一继承**。但本理论也有一张牌：公理E恰恰是**拒绝**纯功能主义的（体验在执行里，不在功能结构里）。本立场是"功能主义 + 执行唯实论"的混合体。

- **Relation to functionalism:** this conjecture restates, at the level of life, the functionalism of philosophy of mind. Functionalism has its own unsolved problems (multiple realizability, the "absence" of qualia, etc.). **This inheritance must be explicitly acknowledged.** But the theory also has a card: Axiom E precisely **rejects** pure functionalism (experience is in execution, not in functional structure). The position here is a hybrid of "functionalism + execution realism."

猜想14专属边界 / Conjecture-14-specific boundaries:

- "闭合"的判定门槛: $d \geq 2$ 是判定"闭合"的操作性门槛, 但 d 本身的定量度量仍是开放的 (见上"闭合深度缺操作度量")。这一门槛目前是**结构性的**, 不是**可测量的**。

- **The threshold of "closure":** $d \geq 2$ is the operational threshold for "closure," but the quantitative measure of d itself remains open (see "closure depth lacks an operational measure" above). This threshold is currently **structural**, not **measurable**.

- **本质问题 vs 历史问题:** 本猜想回答的是"生命是什么", 不是"生命为什么是这种形态"。它不解释: 为什么地球上有碳基生命而不是硅基生命 (化学约束); 为什么生命演化出了 30 多亿年才出现神经系统 (复杂性的路径依赖); 为什么有些生命形式借用了别人的闭环而活 (寄生的可能性)。不能把历史问题也揽过来。

- **Essential vs historical questions:** this conjecture answers "what life **is**," not "why life **takes this particular form**." It does not explain: why Earth has carbon-based life rather than silicon-based (chemical constraint); why life took over 3 billion years to evolve a nervous system (path-dependency of complexity); why some life forms live by borrowing others' loops (the possibility of parasitism). **Historical questions must not be absorbed.**

- **与 RNA 世界假说的关系:** 本猜想把 RNA 世界从"起源事件"降级为"起源事件的物理实现"。这需要进一步论证——主流生物学界对 RNA 世界的许多具体机制已有共识, 本猜想必须证明它**兼容**这些共识, 而不是**替代**它们。

- **Relation to the RNA-world hypothesis:** this conjecture demotes the RNA world from "origin event" to "physical realization of the origin event." This requires further argument — mainstream biology has consensus on many specific mechanisms of the RNA world, and this conjecture must show it is **compatible** with these, not **replacing** them.

- **最早生命的经验重建:** 本猜想预测最早的生命形态是"极其简单的自催化网络", 但这一预测尚待经验证据 (化石记录、系统发生学推断) 的支持。

- **Empirical reconstruction of the earliest life:** this conjecture predicts that the earliest life forms were "extremely simple autocatalytic networks," but this prediction awaits empirical support (fossil records, phylogenetic inference).

版本层 / Version level:

- **猜想12的撤回:** 本版本 (v1.1) 暂时撤回猜想12 (写入-存储分离) 及绑定的P6、P7。撤回是方法论的, 不是内容上的否认。编号12保留占位。

- **Withdrawal of Conjecture 12:** this version (v1.1) temporarily withdraws Conjecture 12 (Encoding–Storage Dissociation) and its bound predictions P6, P7. The withdrawal is methodological, not a denial of the conjecture's content. Slot number 12 is retained as a placeholder.
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10. 结论 / 10. Conclusion

从三个物理事实出发——生命是耗散结构、信息是物理的、有限资源下的建模必然有损压缩——本理论构造了一台必然的机器：它用历史的有损压缩产物生成预测，接收输入，计算偏差，执行响应，再把执行结果压缩回档案，影响明天的预测。

Beginning from three physical facts — life is a dissipative structure, information is physical, and modeling under finite resources is necessarily lossy — this theory constructs a machine that is inevitable: it generates predictions from its history's lossy-compressed products, receives input, computes deviation, executes a response, and then compresses the result back into the archive, modulating tomorrow's prediction.

在这台机器上，本理论完成了以下安放：

On this machine, the theory accomplishes the following placements:

- **意识的位置：** 不在输入、不在预测、不在偏差、不在结构、不在参数，在当帧执行。
- **The location of consciousness:** not in input, not in prediction, not in deviation, not in structure, not in parameter — in the current-frame execution.
- **时间之矢：** 压缩不可逆，与热力学箭头同源。
- **The arrow of time:** compression is irreversible, and shares its source with the thermodynamic arrow.
- **自我：** 永远是自我的不透明模型，内省找不到裸自我是架构必然。
- **The self:** always an opaque model of the self; the failure of introspection to find a bare self is architecturally necessary.
- **感受质四特征：** 私密、不可言说、直接、简单，全部猜想化。
- **The four features of qualia:** private, ineffable, direct, simple — all rendered as corollaries.
- **神秘体验：** 架构的确定性输出。
- **Mystical experience:** a deterministic output of the architecture.

- **发现：** A 是感受的内容， E 是感受的存在，其本质是系统在自己的生存模型中照见自己偏离自身的瞬间。
- **Discovery:** A is the content of experience, E is the existence of experience, and its essence is the system catching sight, within its own survival model, of the moment it diverges from itself.
- **自由意志：** 开放性 × 自我构成 × 审思效力，审议淘汰不挑选，骰子在集内掷。
- **Free will:** openness × self-constitution × deliberative efficacy; deliberation eliminates rather than selects; the die is cast only within the set.
- **困难问题：** 不被解决而被变换，元猜想证明它不能以命题形式被任何运行系统包含。
- **The hard problem:** not solved but transformed; the meta-conjecture proves it cannot be contained as a proposition within any running system.
- **哲学僵尸：** 概念矛盾+进化不可能的残疾，双重消解。
- **The philosophical zombie:** a conceptual contradiction and an evolutionarily impossible defect — a twofold dissolution.
- **生命的位置：** 生命是信息动力学闭环的首次闭合。不是化学事件，是拓扑事件。物理物质动力学是信息动力学的派生维持子系统。
- **The location of life:** life is the first closure of the information-dynamics loop. Not a chemical event; a topological event. Physical material dynamics is a derived maintenance subsystem of information dynamics.

最后，按本理论自己的最后一本账说一句必须说的话：

Finally, by the theory's own final ledger, one thing must be said:

这篇论文作为文本，是压缩档案；它对世界的全部因果效力，发生在每一个读者理解它的那一次执行里，而那件事，按理论自身，不在这篇论文里，也不可能在这篇论文里。

This paper, as text, is a compression archive; its entire causal efficacy upon the world occurs in the single execution in which each reader understands it — and that event, by the theory's own account, is not in this paper, and cannot be in this paper.

这不是这篇论文的缺陷。这是它论证过的、任何地图都搬不动的那块石头。

This is not a defect of the paper. It is the stone that the paper has argued no map can move.

理论到此为止；理论之外，疆域继续。

The theory ends here; beyond the theory, the territory continues.

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(完 / End)

版本历史 / Version history:

- **v1.0** (初版)：十二条猜想 + 一条元猜想。
- **v1.1** (本版)：新增猜想13（载体派生）、猜想14（生命起源）；暂时撤回猜想12及其绑定的P6、P7和§9边界条目。编号12保留占位。
- **v1.0** (initial): twelve conjectures + one meta-conjecture.
- **v1.1** (this version): adds Conjecture 13 (Carrier-Derivation) and Conjecture 14 (Origin-of-Life); temporarily withdraws Conjecture 12 and its bound predictions P6, P7, and the §9 boundary entry. Slot number 12 is retained as a placeholder.